The c^2 is the speed of light squared—a huge number which, when multiplied by the mass, reminds us how much energy you actually get in this exercise.

Shortly before, during, and after the strong and electroweak forces parted company, the universe was a seething soup of quarks, leptons, and their antimatter siblings, along with bosons, the particles that enable their interactions. None of these particle families is thought to be divisible into anything smaller or more basic, though each comes in several varieties. The ordinary photon is a member of the boson family. The leptons most familiar to the non-physicist are the electron and perhaps the neutrino; and the most familiar quarks are . . . well, there are no familiar quarks. Each of their six subspecies has been assigned an abstract name that serves no real philological, philosophical, or pedagogical purpose, except to distinguish it from the others: *up* and *down*, *strange* and *charmed*, and *top* and *bottom*.

Bosons, by the way, are named for the Indian scientist Satyendra Nath Bose. The word "lepton" derives from the Greek *leptos*, meaning "light" or "small." "Quark," however, has a literary and far more imaginative origin. The physicist Murray Gell-Mann, who in 1964 proposed the existence of quarks as the internal constituents of neutrons and protons, and who at the time thought the quark family had only three members, drew the name from a characteristically elusive line in James Joyce's *Finnegans Wake*: "Three quarks for Muster Mark!" One thing quarks do have going for them: all their names are simple—something chemists, biologists, and especially geologists seem incapable of achieving when naming their own stuff.

Quarks are quirky beasts. Unlike protons, each with an electric charge of +1, and electrons, with a charge of –1, quarks have fractional charges that come in thirds. And you'll never catch a quark all by itself; it will always be clutching other quarks nearby. In fact, the force that keeps two (or more) of them together actually grows stronger the more you separate them—as if they were attached by some sort of subnuclear rubber band. Separate the quarks enough, the rubber band snaps and the stored energy summons $E = mc^2$ to create a new quark at each end, leaving you back where you started.

During the quark–lepton era the universe was dense enough for the average separation between unattached quarks to rival the separation between attached quarks. Under those conditions, allegiance between adjacent quarks could not be unambiguously established, and they moved freely among themselves, in spite of being collectively bound to